eSaral

## 5th Apr 2026 Shift 2 (Actual Paper)

### Mathematics Section A

**1.** Let $\alpha, \beta$ be the roots of the equation $x^2 - x + \mathrm{p} = 0$ and $\gamma, \delta$ be the roots the equation $x^2 - 4x + \mathrm{q} = 0$; $p, q \in \mathbf{Z}$. If $\alpha, \beta, \gamma, \delta$ are in G.P., then $|p+q|$ equals :

(A) 16 (B) 32 (C) 34 (D) 38

**2.** Let $z_1, z_2 \in \mathrm{C}$ be the distinct solutions of the equation $z^2 + 4z - (1 + 12i) = 0$. Then $|z_1|^2 + |z_2|^2$ is equal to :

(A) 18 (B) 22 (C) 29 (D) 34

**3.** If $f : \mathbf{N} \to \mathbf{Z}$ is defined by

$$f(n) = \begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}, k \in N,$$

and $\sum_{n=1}^{k} f(n) = 98$, then $k$ is equal to:

(A) 3 (B) 4 (C) 5 (D) 6

**4.** Let M be a $3 \times 3$ matrix such that

$\mathrm{M}\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix}, \mathrm{M}\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\1\\2\end{pmatrix}$ and $\mathrm{M}\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\1\end{pmatrix}$. If $\mathrm{M}\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1\\7\\11\end{pmatrix}$, then $x + y + z$ equals :

(A) 4 (B) 5 (C) 7 (D) 11

**5.** If the sum of the first 10 terms of the series $\frac{1}{1+1^4 \times 4} + \frac{2}{1+2^4 \times 4} + \frac{3}{1+3^4 \times 4} + \frac{4}{1+4^4 \times 4} + \ldots\ldots$ is $\frac{m}{n}, \gcd(m, n) = 1$, then $m + n$ is equal to :

(A) 256 (B) 264 (C) 276 (D) 284

**6.** Let $A_1, A_2, A_3, \ldots\ldots\ldots, A_{39}$ be 39 arithmetic means between the numbers 59 and 159. Then the mean of $\mathrm{A}_{25}$, $\mathrm{A}_{28}$, $\mathrm{A}_{31}$ and $\mathrm{A}_{36}$ is equal to:

(A) 129 (B) 136 (C) 131.50 (D) 134

**7.** The coefficient of $x^2$ in the expansion of $\left(2x^2 + \frac{1}{x}\right)^{10}, x \neq 0$, is :

(A) 3240 (B) 3360 (C) 3480 (D) 3600

**8.** The probabilities that players A and B of a team are selected for the captaincy for a tournament are 0.6 and 0.4 , respectively. If $A$ is selected the captain, the probability that the team wins the tournament is 0.8 and if B is selected the captain, the probability that the team wins the tournament is 0.7 . Then the probability, that the team wins the tournament, is :

(A) 0.74 (B) 0.76 (C) 0.72 (D) 0.78

**9.** A box contains 5 blue, 6 yellow and 4 red balls. The number of ways, of drawing 8 balls containing at least two balls of each colour, is :

(A) 4100 (B) 4140 (C) 4230 (D) 4290

**10.** A variable X takes values $0, 0, 2, 6, 12, 20, \ldots, \mathrm{n}(\mathrm{n}-1$ with frequencies ${}^{\mathrm{n}}\mathrm{C}_0, {}^{\mathrm{n}}\mathrm{C}_1, {}^{\mathrm{n}}\mathrm{C}_2, {}^{\mathrm{n}}\mathrm{C}_3, {}^{\mathrm{n}}\mathrm{C}_4, {}^{\mathrm{n}}\mathrm{C}_5, \ldots, {}^{\mathrm{n}}\mathrm{C}_{\mathrm{n}}$ respectively. If the mean of this data is 60 , then its median is :

(A) 56 (B) 42 (C) 72 (D) 90

**11.** Let the point P be the vertex of the parabola $y = x^2 - 6x + 12$. If a line passing through the point P intersect the circle $x^2 + y^2 - 2x - 4y + 3 = 0$ at the points R and S , then the maximum value of $(\mathrm{PR} + \mathrm{PS})^2$ is :

(A) 10 (B) 20 (C) 25 (D) 5

**12.** Let the directrix of the parabola $\mathrm{P} : y^2 = 8x$, cut $x$-axis a the point A . Let $\mathrm{B}(\alpha, \beta), \alpha > 1$, be a point on P such tha the slope of $AB$ is $3/5$. If $BC$ is a focal chord of $P$, the six times the area of $\Delta ABC$ is :

(A) 80 (B) 160 (C) 174 (D) 192

**13.** Let the eccentricity e of a hyperbola satisfy the equation $6\mathrm{e}^2 - 11\mathrm{e} + 3 = 0$. If the foci of the hyperbola are $(3, 5$ and $(3, -4)$, then the length of its latus rectum is :

(A) 11/3 (B) 17/3 (C) 15/2 (D) 17/2

**14.** Let a triangle PQR be such that P and Q lie on the lin $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$ and are at a distance of 6 units fron $\mathrm{R}(1, 2, 3)$. If $(\alpha, \beta, \gamma)$ is the centroid of $\Delta\mathrm{PQR}$, then $\alpha + \beta + \gamma$ is equal to :

(A) 4 (B) 5 (C) 6 (D) 8

**15.** If the distance of the point $(a, 2, 5)$ from the image of th point $(1, 2, 7)$ in the line $\frac{x}{1} = \frac{y-1}{1} = \frac{z-2}{2}$ is 4 , then th sum of all possible values of $a$ is equal to :

(A) 11 (B) 9 (C) 6 (D) 4

**16.** Let O be the origin, $\overrightarrow{\mathrm{OP}} = \vec{a}$ and $\overrightarrow{\mathrm{OQ}} = \vec{b}$. If R is th point on $\overrightarrow{\mathrm{OP}}$ such that $\overrightarrow{\mathrm{OP}} = 5\overrightarrow{\mathrm{OR}}$, and M is the poin such that $\overrightarrow{OQ} = 5\overrightarrow{RM}$, then $\overrightarrow{PM}$ is equal to :

(A) $\frac{1}{5}(\vec{a} - 4\vec{b})$ (B) $\frac{1}{5}(\vec{b} - 4\vec{a})$

(C) $\frac{1}{5}(-\vec{a} + 4\vec{b})$ (D) $\frac{1}{5}(-\vec{b} + 4\vec{a})$

**17.** Let $f(x) = \lim_{y \to 0} \frac{(1-\cos(xy))\tan(xy)}{y^3}$. Then the number o solutions of the equation $f(x) = \sin x, x \in \mathbf{R}$ is :

(A) 0 (B) 2 (C) 3 (D) 1

**18.** Let $(2^{1-\mathrm{a}} + 2^{1+\mathrm{a}}), f(\mathrm{a}), (3^{\mathrm{a}} + 3^{-\mathrm{a}})$ be in A.P. and $\alpha$ b the minimum value of $f(\mathrm{a})$. Then the value of the integra $\int_{\log_e(\alpha-1)}^{\log_e(\alpha)} \frac{dx}{(e^{2x} - e^{-2x})}$ is:

(A) $\frac{1}{2}\log_e\left(\frac{4}{3}\right)$

(B) $\frac{1}{4}\log_e\left(\frac{4}{3}\right)$

(C) $\frac{1}{2} \log_e \left(\frac{8}{5}\right)$

(D) $\frac{1}{4} \log_e \left(\frac{8}{5}\right)$

**19.** Let $f : [1, \infty) \rightarrow \mathbf{R}$ be a differentiable function defined as $f(x) = \int_1^x f(\mathrm{t})\mathrm{dt} + (1 - x)(\log_{\mathrm{e}} x - 1) + \mathrm{e}$. Then the value of $f(f(1))$ is :

(A) $(1 + \mathrm{e}^{\mathrm{e}})$

(B) $(1 + \mathrm{e})$

(C) $(1 + \mathrm{e} + \mathrm{e}^{\mathrm{e}})$

(D) $1 + 2\mathrm{e}$

**20.** Let $f(x)$ and $g(x)$ be twice differentiable functions satis fying $f''(x) = g''(x)$ for all $x \in \mathbf{R}, f'(1) = 2g'(1) = 4$ an $g(2) = 3f(2) = 9$. Then $f(25) - g(25)$ is equal to :

(A) 20 (B) 40 (C) -20 (D) -40

## Mathematics Section B

**21.** Let $A = \{1, 4, 7\}$ and $B = \{2, 3, 8\}$. Then the number of elements, in the relation $R = \{((a_1, b_1), (a_2, b_2)) \in ((A \times B) \times (A \times B)) : a_1 + b_2$ divides $a_2 + b_1\}$ is ____ .

**22.** From the point $(-1, -1)$, two rays are sent making angles of $45^\circ$ with the line $x + y = 0$. These rays get reflected from the mirror $x + 2y = 1$. If the equations of the reflected rays are $\mathrm{a}x + \mathrm{b}y = 9$ and $cx + dy = 7$, $a, b, c, d \in \mathbf{Z}$, then the value of $ad + bc$ is ____。

**23.** If $S = \{\theta \in [-\pi, \pi] : \cos\theta \cos\frac{5\theta}{2} = \cos 7\theta \cos\frac{7\theta}{2}\}$, the $n(S)$ is equal to ____ .

**24.** Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x, x \in \mathbf{R}$. Let the maximum value of on $\mathbf{R}$ be $\alpha$. If the area of the region bounded by th curves $g(x) = x^2$ and $h(x) = \beta x^3, \beta > 0$, is $\alpha^2$, then $30\beta$ is equal to ____ .

**25.** Let $y = y(x)$ be the solution of the differential equatio $(\tan x)^{1/2}\, \mathrm{d}y = (\sec^3 x - (\tan x)^{3/2} y)\, \mathrm{d}x, 0 < x < \frac{\pi}{2}, y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5}$. If $y\left(\frac{\pi}{3}\right) = \frac{4}{5}\alpha$, then $\alpha^4$ equals ____ .

## Physics Section A

**26.** Match List - I with List - II.

| List - I | | List - II | |
|---|---|---|---|
| (a) | Meter (L) | (I) | $\sqrt{\frac{hc}{G}}$ |
| (b) | Second (S) | (II) | $\sqrt{\frac{Gh}{c^5}}$ |
| (c) | Kilogram (M) | (III) | $\sqrt{\frac{K^2L^2c^3}{Gh}}$ |
| (d) | Kelvin (K) | (IV) | $\sqrt{\frac{Gh}{c^3}}$ |

where h (Planck's constant), G (gravitational constant) and c (speed of light in vacuum) as fundamental units. Choose the correct answer from the options given below :

(A) A-II, B-IV, C-I, D-III (B) A-IV, B-II, C-I, D-III

(C) A-IV, B-I, C-II, D-III (D) A-III, B-I, C-II, D-IV

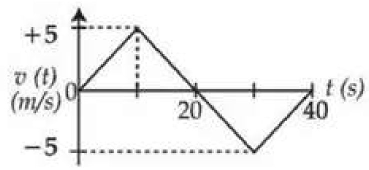
**27.** In an experiment to determine the resistance of a given wire using Ohm's law, the voltmeter and ammeter readings are noted as 10 V and 5 A , respectively. The least counts of voltmeter and ammeter are 500 mV and 200 mA , respectively. The estimated error in the resistance measurement is ____ $\Omega$

(A) 0.25 (B) 2 (C) 2.5 (D) 0.18

**28.** A mass of 1 kg is kept on a inclined plane with $30^\circ$ inclination with respect to horizontal plane and it is at rest initially. Then the whole assembly is moved up with constant velocity of $4\ \mathrm{m/s}$. The work done by the frictional force in time 2 s is ____ J. (Take $g = 10\ \mathrm{m/s^2}$ )

(A) 20 (B) 25 (C) 30 (D) 10

**29.** The velocity $(v)$ versus time $(t)$ plot of a particle i shown in the figure, for a time interval of 40 s . The tota distance travelled by the particle and the average velocit during this period are, respectively ____ .




(A) 25 m and zero

(B) 50 m and zero

(C) 100 m and zero

(D) 100 m and $2.5\ \mathrm{m/s}$

**30.** A wheel initially at rest is subjected to a uniform angula acceleration about its axis. In the first 2 s it rotate through an angle $\theta_1$ and in the next 2 s it rotates throug an angle $\theta_2$. The ratio $\frac{\theta_2}{\theta_1}$ is ____ .

(A) 6 (B) 3

(C) 4 (D) $\frac{1}{3}$